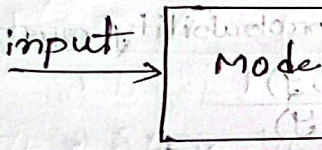
Lecture – 04.
Saturday ☺
(12/09/26)

Topic : Naive Bayes.

i) $P(a/b) = \frac{P(a,b)}{P(b)}$

ii) $P(a,b) = P(a/b)\,P(b)$

Model কে আগে train করা লাগবে।



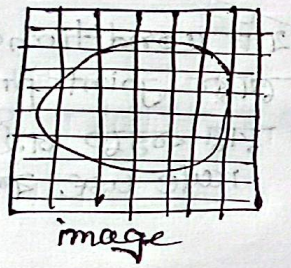


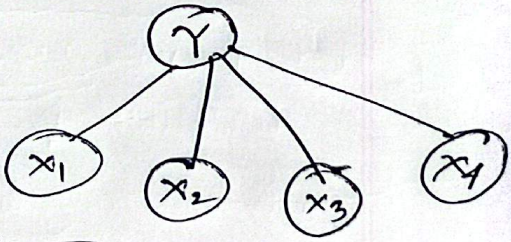
image এ pixel grids থাকবে।
pixel এর corresponding value থাকবে।

— x —

Data এর feature থাকবে এবং label থাকবে।
↓ ↓
characteristics cat বা Dog

x_1	x_2	x_3	x_4	Y
height	weight	tail	shape	cat/Dog

features. label



features গুলো label এর উপর dependent.

$x_1 = a,\ x_2 = b$
$Y = ?$

New data দিয়েছি।
এটা যেটা কোন ক্লাসের বলে Classification.

$P(Cat | x_1 = a, x_2 = \ldots\ldots) = (0.2)$
$P(Dog | x_1 = a, x_2 = \ldots\ldots) = (0.8)$ ←

যতগুলো label থাকবে সবগুলোর জন্য probability calculate করবে। যার probability বেশি তাকে accept করবে।

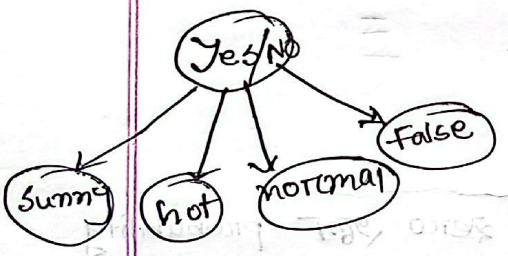
conditional probability ← $P(cat | x_1, x_2, x_3) = \dfrac{P(cat, x_1, x_2, x_3 \ldots)}{P(x_1, x_2, x_3 \ldots\ldots)}$

Example: Slide (weather, golf playing এর টেবিল).

golf → [Yes / NO]

$P(Yes / sunny, hot, Normal, False) \rightarrow \dfrac{P(yes, sunny, hot, normal, false)}{P(sunny, hot, normal, false)}$ ← joint probability

$P(NO / sunny, hot, Normal, False)$



যার parent নেই সে direct বসবে।

$= P(yes)\, P(sunny/yes)\, P(hot/yes)\, P(normal/yes)\, P(false/yes)$

$= \dfrac{\left(\frac{9}{14}\right)\cdot\left(\frac{3}{9}\right)\cdot\left(\frac{2}{9}\right)\cdot\left(\frac{6}{9}\right)\cdot\left(\frac{6}{9}\right)}{P(sunny, hot, Normal, false)}$

$= \dfrac{(\quad)}{P(\ldots\ldots)} = \dfrac{0.021}{P(S, h, N, F)}$

$= \dfrac{0.021}{0.021 + 0.005}$

CamScanner

$$P(NO/sunny, hot, Normal, False)$$

$$= \frac{P(NO, sunny, hot, Normal, False)}{P(sunny, hot, Normal, False)}$$

$$= \frac{P(NO) \cdot P(sunny/NO) \cdot P(hot/NO) \cdot P(Normal/NO) \cdot P(False/NO)}{P(sunny, hot, Normal, False)}$$

$$= \frac{5/14 \times 2/5 \times 2/5 \times 1/5 \times 2/5}{P(sunny, hot, Normal, False)} = \frac{0.005}{P(S, H, N, F)} = \frac{0.005}{0.021 + 0.005}$$

classify করতে বলতেছে কে খেলবে কি খেলবে না।

Decision আমি এখান থেকেই নিতে পারছি।

যদি probability জানতে চায় তাহলে further calculate করা লাগবে।

$$P(S, H, N, F) = P(yes, S, H, N, F) + P(NO, S, H, N, F)$$

$$= 0.021 + 0.005$$

$$=$$

exactly probability বের করতে বলতেছে।

"Smoothing"

Label এর against এ feature এর value zero হলে probability zero হয়ে যায় যা হওয়া। এটা এড়াতে Smoothing করা হয়।

- খুব শেষেই চিন্তা করবো feature এর value ও zero এবে না।
- Smoothing factors Question এ বলা থাকবে। ($k = 1, 2 \cdots$)

Sat CT
Topic : CSP ☺

Lecture – 05.
15/09/2024.
(Tuesday).

Smoothing কেন দরকার? Smoothing বড় গ্রুপ কি হবে? [theory আসতে পারে]

✓ Zero frequency problem

Example

$K = 2$

$$P(\text{Yes} \mid S, H, N, F) = \frac{P(\text{yes}, S, H, N, F)}{P(S, H, N, F)} = \frac{P(\text{yes}), P(S/\text{yes}), P(H/\text{yes}), P(N/\text{yes}), P(F/\text{yes})}{P(S, H, N, F)}$$

$$P(\text{NO} \mid S, H, N, F) = \frac{P(\text{NO}, S, H, N, F)}{P(S, H, N, F)}$$

Yes ~~NO~~ NO

$\frac{9+2}{14+2*2}$ $\frac{5+2}{14+2*2}$

$$= \frac{P(\text{NO}), P(S/\text{NO}), P(H/\text{NO}), P(N/\text{NO}), P(F/\text{NO})}{P(S, H, N, F)}$$

$$= \frac{\frac{7}{18} \times \frac{4}{5+3\times2} \times \frac{4}{5+3\times2} \times \frac{3}{5+2\times2} \times \frac{4}{5+2\times2}}{P(S, H, N, F)}$$

$$\frac{11/18 \times \frac{5}{9+3*2} \times \frac{4}{9+3*2} \times \frac{8}{9+2*2} \times \frac{8}{9+2*2}}{P(S, H, N, F)}$$

Topic : Fuzzy and Crisp Logic

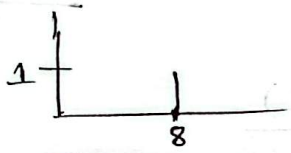
Fuzzy → between (0 ~ 1) anything.

Crisp → True or false

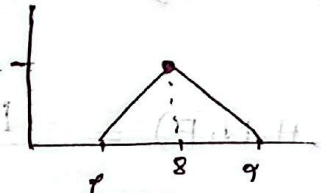
Fuzzy → Degree of membership.

Fuzzy → already ঘটেনা জানা

8 → 1
8 না হলে → 0



Crisp



Fuzzy

According to fuzzy logic:

Given $A = (0.2, 0.5, 0.4)$

$B = (0.5, 0.3, 0.7)$

$\therefore A \cup B = \max(A \cup B)$

$= (0.5, 0.5, 0.7)$

$A \cap B = \text{minimum}(A \cap B)$

$= (0.2, 0.3, 0.1)$

$\overline{A} = 1 - A$

$= (0.8, 0.5, 0.6)$

$A - B = \min(A) - \text{complement of } B \quad \min(\mu(A) - 1 - \mu(B))$

$= (0.2, 0.5, 0.3)$

A = (1, 0.5, 0.3, 0.2)

B = (0.5, 0.7, 0.2, 0.1)

∴ A∪B = (1, 0.7, 0.3, 0.4)

= 1/2 + 0.7/3 + 0.3/1 + 0.4/5

A∩B = (0.5, 0.5, 0.2, 0.2)

$\overline{A}$ = (0, 0.5, 0.7, 0.8)

Fuzzification Method → normal value কে membership num এতে express করা : i) singleton ii) Gaussian, iii) Triangulare, iv) Trapezoidal.

lecture – 06. "Saturday"

Fuzzy linguistic variables :-

AND, OR, NOT

↑

operations + fuzzification Method

Lecture – 07. (22/09/2026) (Tuesday)

→ Fuzzy subset :

A = (0.5, 0.8, 0.3)

B = (0.6, 0.7, 0.9)

↓ True ↓ False ↓ True.

যদি A ⊆ B হয় তাহলে A ⊂ B.

individual subset বের করবো

∴ A ⊆ B = (True, false, True)

→ Arithmetic operation : A = (-2, 5, 8)

B = (5, 9, 10)

∴ $\frac{10+15}{2}$ = 12.5 (average spread)

8 − (−2) = 10 ⌐

10 − (−5) = 15 ⌐ } কতটুকু spread

∴ A+B = (~~11-12.5~~, 11, 12.5+11)

Step 1: first এ variance calculate করবো

Step 2: variance এর average নিবো

Step 3: তারপর A ও B এর middle value এর যোগফল নিবো বা average নিবো.

Step 4: middle e value থেকে ~~var~~ average variance ~~এ~~ ~~এ~~ first num থেকে minus এবং last num এর সাথে plus.

Topic : First_order logic

Constants : specific names or things / objects.

Variables: Specific কোন person না, It could be anything.

Predicates: Actions/relations → student(Rafid)
↳ True / False দেয়।

Functions: Mappings from one object to another.

Fatherof(John) = David.

↳ output হিসাবে আরেকটা object দেয়।

Quantifiers :

Universal (∀) → for all

Existential: (∃) → at least one (একটার জন্য true হবে)

Conjuction: and implication: (→) if.... then

Disjunction: or

Biconditional : $A \leftrightarrow B = (A \rightarrow B,\ B \rightarrow A)$

Negation : $(\neg) \rightarrow$ NOT

**

i) Rule
ii) fact
iii) Conclusion.

problem 1

✓ All humans are Mortal.

$= \forall x\ (Human(x) \rightarrow Mortal(x))$ } rule

✓ Socrates is a human.

$= Human(Socrates)$

$x = Socrates$

$Human(Socrates) \rightarrow Mortal(Socrates)$

(Human(Socrates): True)

Conclusion : $Mortal(Socrates)$.

problem 2

$\forall x\ (Student(x) \rightarrow \exists y\ studies(x, y)) \Rightarrow$ rule

$Student(Rahim) \Rightarrow$ fact

Sol^n: fact থেকে x এর value টা বের করা।

$x =$ Rahim.

$Student(Rahim) \rightarrow \exists y\ studies(Rahim, y)$

(Student(Rahim): True)

Conclusion : $\exists y\ studies(Rahim, y) =$ true (Answer)

problem 3

$\forall x \forall y$ (Fathere. (y, x) $\rightarrow$ Parent (y, x))

Fathere. (Ali, Rahim)

Soln:

y = Ali
x = Rahim

fathere (Ali, Rahim) $\rightarrow$ parent (Ali, Rahim)
True

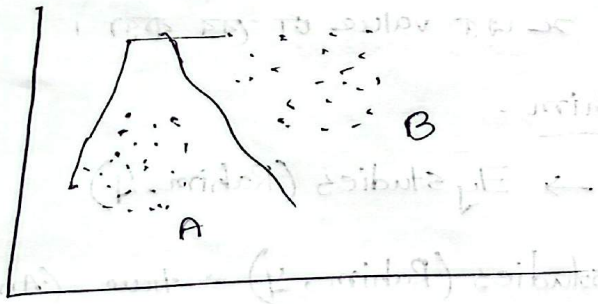
conclusion : parent (Ali, Rahim) = True

Next sat
CT → topic
naive bayes
+
k-means.

problem 1

lecture – 08
Sat (26-09-26)

Topic: Unsupervised Learning
→ K-means clustering



centroid → reference point

Classify the following data into two clusters using the k-means algorithm (3,4) (6,7) (4,5) (5,7) (2,6)

Soln: Iteration 1
K = 2

centroid দেয়া না থাকলে যেকোনো ২টাকে ধরে নিবো।

Let, class -1 : (3,4)

class -2 : (4,5)

	class-1 (3,4)	class-2 (4,5)	
(3,4)	0	1.414	class-1
(6,7)	4.242	2.828	class-2
(4,5)	1.414	0	class-2
(5,7)	3.605	2.236	class-2
(2,6)	2.236	2.236	class-1

$\sqrt{(3-4)^2+(4-5)^2}$

∴ class-1 : (3,4)(2,6) $= \left(\frac{3+2}{2}, \frac{4+6}{2}\right) = \left(\frac{5}{2}, \frac{10}{2}\right) = (2.5, 5)$

class-2 : (6,7) (4,5) (5,7) $= \left(\frac{6+4+5}{3}, \frac{7+5+7}{3}\right)$

$= (5, 6.33)$

Iteration 2

New centroid,

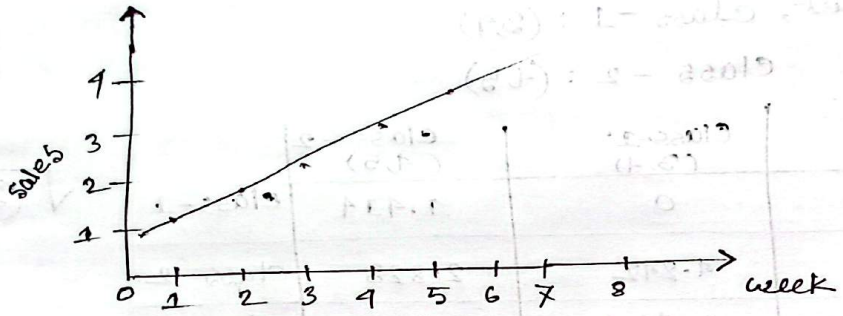
class-1 : (2.5, 5)

class-2 : (5, 6.33)

→ stable না হওয়া পর্যন্ত iteration চালাবো।
→ stable হলে Algorithm stop.

Topic :- Regression → কোনো কিছু prediction করা।
→ linear regression (m, c বের করা)
→ logistic regression → probability হিসেবে output দেবো

Linear Regression :



$y = mx + c$

$m = \frac{\overline{xy} - \bar{x}\bar{y}}{\overline{x^2} - \bar{x}^2} = 0.66$ (বার বলতে average)

$x = 0$ হলে $y = ?$ ← $c = \bar{y} - m\bar{x} = 0.54$

$\therefore \boxed{y = 0.66x + 0.54}$ → regression equation.

7-th week এর sales কত? → simply $x = 7$ বসাবো regression equation এ।

Logistic Regression : → Sigmoid function.

$z = mx + c$ (linear equation)

Sigmoid function equation : $S(z) = \frac{1}{1+e^{-z}}$

↑ [0, 1] এর মাঝখানে যেকোনো value হতে পারে।

1) calculate the probability of pass for the student who studied 30 hours.

given,

$z = -64 + 2x$

$= -64 + 2 \times 33$

$= 2$

$\therefore S(z) = \frac{1}{1+e^{-2}} = 0.88$

Threshold value always 0.5

✓ z এর equation দেওয়া থাকবে।

② $0.95 = \frac{1}{1+e^{-z}}$

$\Rightarrow z = (2.94)$

$\therefore z = -64 + 2x$

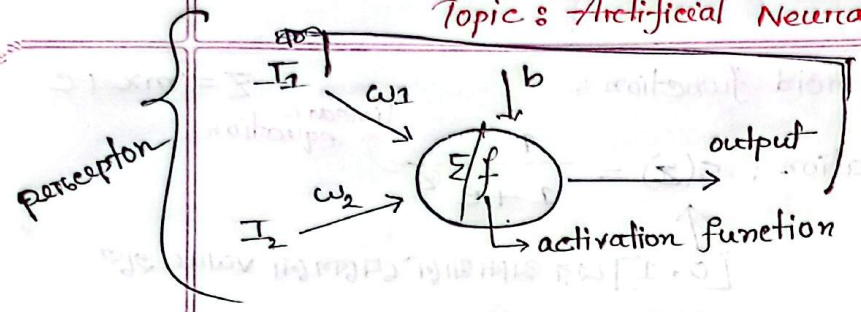
$\Rightarrow 2.94 = -64 + 2x$

$\Rightarrow x = \frac{2.94 + 64}{2}$

$\Rightarrow x = 33.47$ hours.

Lecture – 09.
29/06/26.

Topic : Artificial Neural Network (ANN) (Tuesday)



w = weight (কোন input টার priority বেশি তা decide করে)

b = bias.

I = input

training = forward + backward propagation -

learning rate (η)
↳ ধ্রুবক

problem 1 : d = expected output.

first round →

Given,

$I_1 = 1, I_2 = 1, d = 1$

$w_1 = 0.5, w_2 = 0.3, b = 0.1$

$\eta = 0.1$

sol^n: weighted sum, $v = I_1w_1 + I_2w_2 + b$

$= 1 \times 0.5 + 1 \times 0.3 + 0.1$

$= 0.9$

sigmoid

$\frac{1}{1+e^{-v}}$

$f = \frac{1}{1+e^{-0.9}} = 0.71$ (calculated output) [actual output 1 এই value তে adjust করা লাগবে]

বস্তুত adjustment দরকার

$\delta = f(1-f)(d-f)$ (ডেল্টা)

$= 0.71(1-0.71)(1-0.71)$

$= 0.059$

এবার নতুন weight বের করবো,

$w_1 = w_1 + \Delta w_1 = 0.5 + \eta \times \delta \times I_1 = 0.5 + 0.1 \times 0.059 \times 1 = 0.5059$

$w_2 = w_2 + \Delta w_2 = 0.3 + \eta \times \delta \times I_2 = 0.3 + 0.1 \times 0.059 \times 1 = 0.3059$

$b = b + \Delta b = 0.1 + \eta \times \delta = 0.1 + 0.1 \times 0.059 = 0.105$

CamScanner

updated weight use করবো 2nd input-এর জন্য।

Second round,

$I_1 = 0, I_2 = 0, d = 0$

$w_1 = 0.5059, w_2 = 0.3059, b = 0.1059, \eta = 0.1$

$v = I_1 w_1 + I_2 w_2 + b$

$= 0 \times 0.5059 + 0 \times 0.3059 + 0.1059 = 0.1059$

$f = \frac{1}{1+e^{-0.1059}} = 0.526$

$\delta = f(1-f)(d-f)$

$= 0.526 (1 - 0.526) (0 - 0.526)$

$= -0.1311$

$w_1 = w_1 + \Delta w_1 = 0.5059 + \eta \times \delta \times I_1 = 0.5059$ (0)

$w_2 = w_2 + \Delta w_2 = 0.3059 + \eta \times \delta \times I_2 = 0.3059$ (0)

$b = b + \Delta b = 0.1059 + \eta \times \delta = 0.09279$

Testing করবো,

$I_1 = 0, I_2 = 1, d = ?$

$w_1 = 0.5059, w_2 = 0.3059, b = 0.09279$ (updated value)

$v = 0 \times 0.5059 + 1 \times 0.3059 + 0.09279$

$= 0.39869$

$f = \frac{1}{1+e^{-0.39869}} = 0.598$ (Ans)